Employee Attrition and Workload Analysis

Willem Donnelly

This notebook looks at two questions from our HR director:

1. How do employees who left the company compare with employees who stayed?
2. Do employees who work longer hours tend to receive higher performance evaluations?

The analysis uses the full `HR_comma_sep.csv` dataset, where each row is one employee record.

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
from scipy import stats
```

1. Prepare the Data and Measure Attrition

First, I load the dataset and check its size. Then I check for missing values in the four columns this analysis uses: whether the employee left, their satisfaction level, their average monthly hours, and their last evaluation score.

In [2]:
```
df = pd.read_csv("HR_comma_sep.csv")

print(f"Rows: {df.shape[0]:,}")
print(f"Columns: {df.shape[1]}")

# Missing values in the columns used in this analysis
analysis_columns = ["left", "satisfaction_level", "average_montly_hour
print("\nMissing values per column:")
print(df[analysis_columns].isna().sum())
```

```
Rows: 14,999
Columns: 10

Missing values per column:
left                     0
satisfaction_level       0
average_montly_hours     0
last_evaluation          0
dtype: int64
```

In [3]:
```
# Keep the original left column and add a readable label for each grou
df["employment_status"] = df["left"].map({0: "Stayed", 1: "Left"})

status_counts = df["employment_status"].value_counts()
status_table = pd.DataFrame({
    "Count": status_counts,
    "Percentage": (status_counts / len(df) * 100).round(2),
})
status_table
```

Out[3]:

employment_status	Count	Percentage
Stayed	11428	76.19
Left	3571	23.81

Attrition rate: 3,571 of the 14,999 employees in the dataset left the company, an attrition rate of **23.81%**. The denominator is all 14,999 employee records.

2. Compare Employees Who Left With Those Who Stayed

Chosen measure: employee satisfaction (`satisfaction_level`)

Satisfaction is scored from 0 (least satisfied) to 1 (most satisfied). It matters to HR because it's one of the few warning signs a company can measure *before* someone quits. If employees who left were noticeably less satisfied, regular satisfaction surveys could help HR spot people at risk of leaving while there's still time to act.

In [4]:
```
summary = (df.groupby("employment_status")["satisfaction_level"]
             .agg(["count", "mean", "median", "std"])
             .rename(columns={"count": "Count", "mean": "Mean",
                              "median": "Median", "std": "Std. Deviati
             .loc[["Stayed", "Left"]]
             .round(3))
summary
```

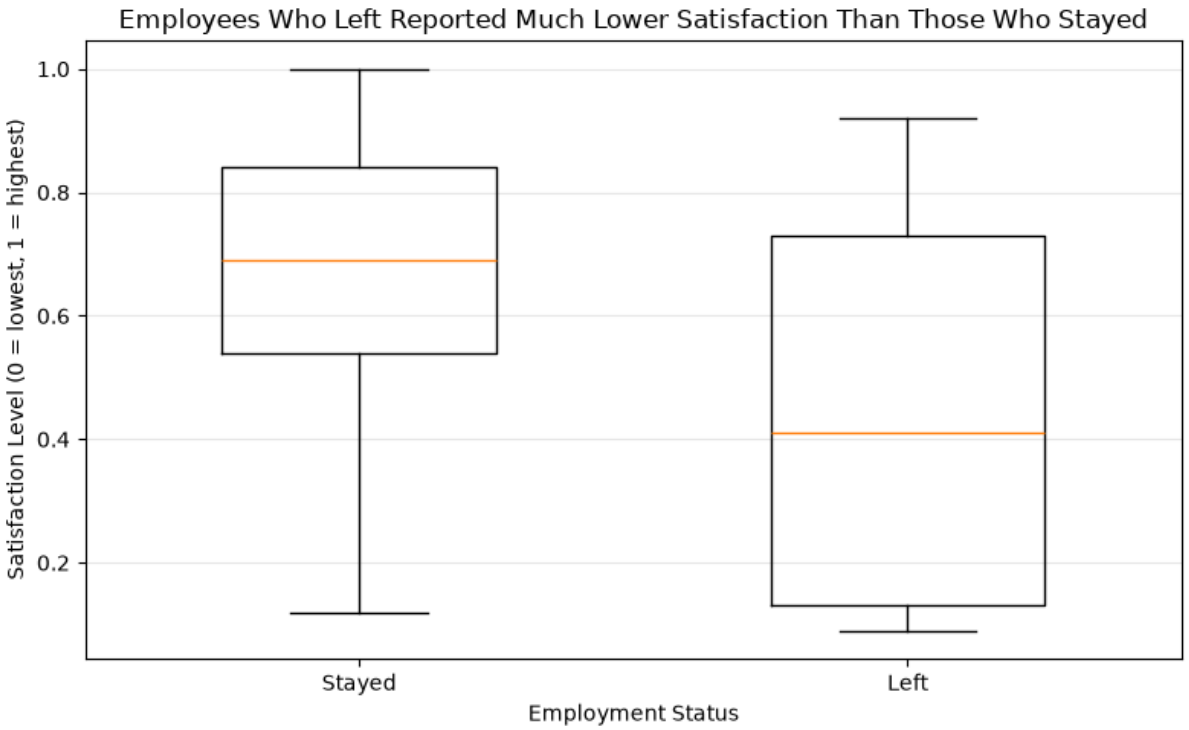
Out[4]:

employment_status	Count	Mean	Median	Std. Deviation
Stayed	11428	0.667	0.69	0.217
Left	3571	0.440	0.41	0.264

In [5]:
```
fig, ax = plt.subplots(figsize=(8, 5))

groups = ["Stayed", "Left"]
data = [df.loc[df["employment_status"] == g, "satisfaction_level"] for
ax.boxplot(data, widths=0.5)
ax.set_xticks([1, 2], groups)

ax.set_title("Employees Who Left Reported Much Lower Satisfaction Than
ax.set_xlabel("Employment Status")
ax.set_ylabel("Satisfaction Level (0 = lowest, 1 = highest)")
ax.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()
```



What the boxplots show: The biggest difference is in the middle of each group. The median satisfaction for employees who stayed was **0.69**, compared with **0.41** for employees who left. The spread differs too. Employees who left are spread across almost the entire scale (the middle half runs from about 0.13 to 0.73), with a large cluster at very low satisfaction. The middle half of the employees who stayed sits in a tighter, higher band, from about 0.54 to 0.84. The boxes do overlap, though, so some employees who left were fairly satisfied, and some who stayed were not. Low satisfaction is a strong pattern, not a perfect predictor.

3. Test the Difference in Means

Null hypothesis (H_0): The average satisfaction level of employees who left is the same as the average satisfaction level of employees who stayed.

Alternative hypothesis (H_1): The average satisfaction level of employees who left is different from the average satisfaction level of employees who stayed.

I use Welch's t-test because it doesn't assume both groups have the same spread. That fits here, since the standard deviations are 0.264 for employees who left and 0.217 for those who stayed.

In [6]:
```
left_sat = df.loc[df["employment_status"] == "Left", "satisfaction_lev
stayed_sat = df.loc[df["employment_status"] == "Stayed", "satisfaction

# Welch's two-sided t-test, Left group first
t_result = stats.ttest_ind(left_sat, stayed_sat, equal_var=False)
mean_difference = left_sat.mean() - stayed_sat.mean()

def format_p(p):
    # Very small p-values can underflow to 0.0 in floating point
    if p == 0:
        return "< 1e-300 (too small for the computer to represent exac
    return f"{p:.2e}"

print(f"Mean satisfaction, Left:   {left_sat.mean():.4f}")
print(f"Mean satisfaction, Stayed: {stayed_sat.mean():.4f}")
print(f"Mean difference (Left - Stayed): {mean_difference:.4f}")
print(f"t statistic: {t_result.statistic:.2f}")
print(f"p-value: {format_p(t_result.pvalue)}")
```

```
Mean satisfaction, Left:   0.4401
Mean satisfaction, Stayed: 0.6668
Mean difference (Left - Stayed): -0.2267
t statistic: -46.64
p-value: < 1e-300 (too small for the computer to represent exactly)
```

Technical interpretation: The t statistic is about −46.64, and the p-value is far below 0.05 (smaller than 1×10^{-300}). We reject the null hypothesis. There is a statistically significant difference in mean satisfaction between employees who left and employees who stayed.

What this means for HR: Employees who stayed had the higher average satisfaction, **0.667**, compared with **0.440** for employees who left. That's a gap of about **0.23** on a 0-to-1 scale, the equivalent of 23 points on a 100-point survey. The gap is far too large to be explained by random chance. In short, people who left the company were, on average, a lot less satisfied than people who stayed.

4. Investigate Hours and Evaluations

This section uses all 14,999 employees, whether they left or stayed, to see whether average monthly hours move together with the most recent evaluation

score.

In [7]:
```
r_result = stats.pearsonr(df["average_montly_hours"], df["last_evaluat

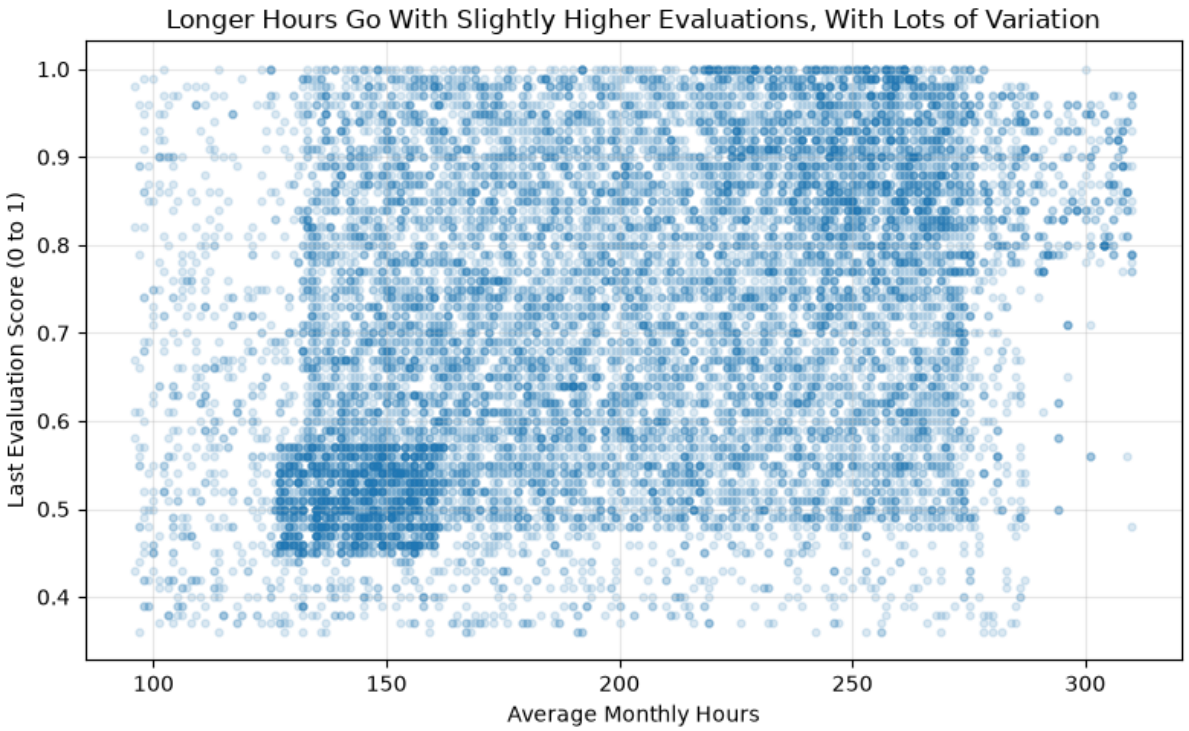
print(f"Pearson correlation (r): {r_result.statistic:.3f}")
print(f"p-value: {format_p(r_result.pvalue)}")
print(f"r squared: {r_result.statistic ** 2:.3f}  "
      f"(share of the variation in evaluations that lines up with hour
```

```
Pearson correlation (r): 0.340
p-value: < 1e-300 (too small for the computer to represent exactly)
r squared: 0.115  (share of the variation in evaluations that lines up
with hours)
```

In [8]:
```
fig, ax = plt.subplots(figsize=(8, 5))

ax.scatter(df["average_montly_hours"], df["last_evaluation"], alpha=0.

ax.set_title("Longer Hours Go With Slightly Higher Evaluations, With L
ax.set_xlabel("Average Monthly Hours")
ax.set_ylabel("Last Evaluation Score (0 to 1)")
ax.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



Do employees who work longer hours tend to receive higher evaluations?
Yes, somewhat. The correlation is **r = 0.34**, a positive but moderate-to-weak relationship. Employees who work more hours *tend* to have somewhat higher evaluations, and the scatterplot shows this as a gradual upward drift from left to right.

Statistical significance: The p-value is far below 0.05, so the relationship is statistically significant. With almost 15,000 employees, though, even a modest pattern will come out significant. Significance tells us the pattern is real, not how *strong* it is.

How useful hours are for one employee: Hours tell us very little about any one person's evaluation. Squaring the correlation ($r^2 \approx 0.12$) shows that hours line up with only about **12%** of the variation in evaluation scores, which leaves about 88% unexplained. The scatterplot makes this clear: at almost every hours level, evaluations range from the 0.4s to 1.0. Someone working 250 hours a month could easily have a lower score than someone working 150. The plot also shows a dense cluster of employees working about 130 to 160 hours with evaluations around 0.45 to 0.57, a group that could be worth a closer look on its own. Hours are one small piece of the picture, not a reliable way to predict how an individual will be evaluated.

5. HR Briefing

Employees who left the company were noticeably less satisfied than those who stayed, with an average satisfaction score of 0.44 compared with 0.67, a gap of about 23 points on a 100-point scale. Longer hours were only modestly connected to better reviews (correlation of 0.34), and hours account for just about 12% of the differences in evaluation scores, so working more hours is far from a guarantee of a strong review. Because this data describes what already happened rather than coming from a controlled experiment, it shows that low satisfaction and leaving go together, but it can't prove that low satisfaction *caused* people to leave. As a next step, HR could look into whether the employees who left with very low satisfaction were also the ones carrying the heaviest workloads, such as the most projects or the longest hours, to see whether burnout is driving attrition.